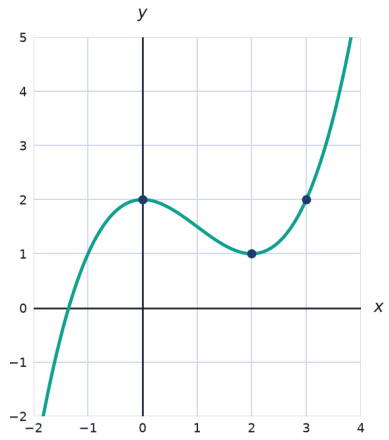
Integers and the number line

Core Concepts: Integers and the number line

1 For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(2)$.



a. 7

b. 9

c. 4

d. 11

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

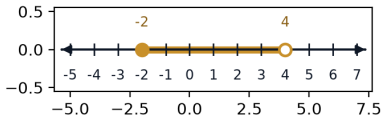
5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$



6 A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(4)$.

a. 11

b. 13

c. 8

d. 15

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8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

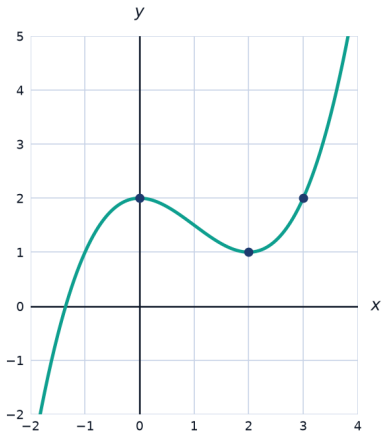
a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

9 A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?



a. 22.5

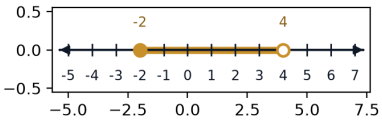
b. 23.5

c. 24.5

d. 26.5

Exam-Style Problem Solving & Modeling

10 For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(5)$.



a. 13

b. 15

c. 10

d. 17

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11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

12 A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

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